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Lossless Compression of Large Language Model-Generated Text via Next-Token Prediction

Yu Mao
MBZUAI
Abu Dhabi
yu.mao@mbzuai.ac.ae

Holger Pirk
Imperial College London
England
hlgr@ic.ac.uk

Chun Jason Xue
MBZUAI
Abu Dhabi
jason.xue@mbzuai.ac.ae

Abstract

As the deployment and utilization of LLMs continue to expand, the volume of LLM-generated data is increasing at an unprecedented rate. Therefore, the need for *effective and lossless compression of LLM-generated data has become increasingly critical in text data management*. However, compressing LLM-generated data poses significant challenges compared to traditional text data, such as human-generated or machine-generated content. Traditional machine-generated data is predominantly derived from computational processes or device outputs, characterized by its highly structured and discrete nature, often limited to low-level information such as labels or numerical values. As a result, conventional lossless compression algorithms are able to exploit this structure to efficiently compress such data. In contrast, the complexity and diversity of LLM-generated data require different approaches to achieve effective lossless compression. To this end, this work conducts the first investigation into effective lossless compression methods for LLM-generated data. Interestingly, since LLMs are trained through next-token prediction, we observe that LLM-generated data exhibits a high predictability for LLMs, which allows LLMs to achieve remarkable lossless compression of LLM-generated data. We conduct extensive experiments with 14 representative LLMs and 8 LLM-generated datasets from different categories. The experimental results reveal that LLM-based prediction methods achieve exceptional compression rates, exceeding 20×, which significantly outperforms the 3x achieved by the state-of-the-art lossless compressor Gzip. Moreover, we demonstrate that the superior performance of LLM-based compressors generalizes across different LLM scales and diverse dataset categories, making them a useful tool in lossless data compression.

1 Introduction

The proliferation of large language models (LLMs) has led to a significant surge in AI-generated text across diverse domains [6, 13, 42, 44]. Given proper instructions from humans, LLMs can generate desirable, high-quality text for multiple purposes. The great capability and versatility of LLMs have made the use and deployment of LLMs continue to expand, which contributes an exponentially growing proportion of the text data in the world. OpenAI's ChatGPT reportedly produces approximately 100 billion words daily [43]. In academia, an estimated 60,000 scholarly articles in 2023—over 1% of all publications—were likely written with LLM assistance [20], with 17.5% of newly published computer science papers and 16.9% of peer review texts incorporating LLM-generated content [28]. An analysis of over 6 billion sentences from Common Crawl further revealed that more than 57% were machine-translated, highlighting the extent of automated content generation [47]. In fact, LLMs have also influenced various fields, including creative writing, healthcare, law, and software development. Authors use LLMs to co-author novels and generate narratives, as seen in Ayad Akhtar's play "McNeal" [1]. In healthcare, they aid in diagnostics, automate documentation, and enhance medical education [50]. In the legal domain, LLMs assist in drafting legal documents and summarizing cases [7]. Software development has also benefited from LLM-powered code generation, streamlining programming workflows [8]. In one of the largest dataset host platforms – Huggingface,[1] we find that the LLM-generated data grows at an exponential speed, as shown in Fig. 1. Furthermore, we fit a quadratic polynomial regression model, which predicts that the LLM-generated data will continue to expand up to 16,000 TB within the next decade. The exponential growth of LLM-generated text poses significant demands and challenges for efficiently storing LLM-generated data.

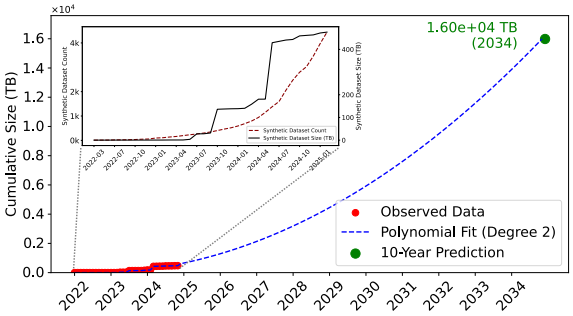

Figure 1: Trend of increasing LLM-generated data.

The rise of LLMs has introduced a new category of machine-generated data: *LLM-generated text*. Prior to the era of LLMs, machine-generated data usually referred to structured texts such as logs [5, 26]. For example, an IoT device can generate up to 1 TB per day in industrial applications, which are typically structured, containing numerical values and categorical information [53]. As machine-generated text grows at a much higher speed than that from humans, a number of traditional compressors are designed to reduce the data storage volume of machine-generated text through the exploitation of the intrinsic structures [52, 56, 57]. In contrast, LLM-generated text shares the trait of even more significantly rapid growth but exhibits distinct characteristics.

Therefore, in this work, we conduct the first investigation of effective lossless compression approaches for LLM-generated data. We begin with an in-depth analysis of the characteristics and compressibility of LLM-generated synthetic datasets. Specifically, we

[1] https://huggingface.co/

consider and curate 3 LLM-generated synthetic datasets from different domains. Based on those datasets, we measure the redundancy levels in LLM-generated text through n-gram statistics (Fig. 2), and find it is challenging to simply deduplicate the LLM-generated text. Furthermore, we assess the compression potential by examining entropy per byte and mutual information (Table 2), which shows that LLM-generated data share similar levels of redundancy and information density to human-generated data. Consequently, we find that traditional compression approaches, such as non-neural and neural compressors, can not achieve satisfactory compression ratios (Table 3). Most methods achieve only a compression ratio of 3–6×, including entropy compression techniques based on neural networks. Our findings provide an empirical foundation for designing efficient compression techniques *tailored to the unique characteristics* of LLM-generated text.

To this end, we propose to incorporate LLMs to compress LLM-generated text. As LLMs are trained through next-token prediction, LLMs demonstrate high predictability on the LLM-generated data. Hence, we can leverage the inherent predictability of LLMs to build powerful lossless compressors for LLM-generated data with remarkable compression capabilities. The need for *compression rather than re-generation* arises from the randomness in LLM generation— even with the same prompt and model, exact re-generation remains infeasible. Therefore, an LLM can serve as a strong predictor for other LLM-generated data, but not with absolute accuracy. We conduct extensive experiments using 14 representative LLMs and 8 LLM-generated datasets spanning diverse categories. The results demonstrate that LLM-based prediction methods achieve exceptional compression rates, exceeding 20×, which significantly surpasses the 3× compression ratio achieved by Gzip, the state-of-the-art lossless compression algorithm. Furthermore, we show that the superior performance of LLM-based compressors generalizes across various LLM scales and dataset categories, highlighting their robustness and potential as a valuable tool for lossless data compression. Finally, we also conduct an in-depth analysis of factors influencing compression efficiency, such as chunk size and data scale.

To summarize, this paper makes the following contributions:

- To the best of our knowledge, we present the *first systematic analysis* of the characteristics of LLM-generated data in terms of compressibility. As an emerging new category of machine-generated data, LLM-generated data exhibits significantly different characteristics from traditional machine-generated data while being similar to human-generated data. Consequently, traditional compressors can not effectively compress LLM-generated data.
- We are also the first to propose leveraging LLMs to compress LLM-generated synthetic data. We show that LLMs have high inherent predictability for the LLM-generated data, inherited from the training with next-token prediction, despite differences in the training sources. It allows LLMs to compress the LLM-generated data effectively.
- We adopt 8 datasets from different domains, and 14 state-of-the-art LLMs of different scales, trained on diverse data sources, and demonstrate that LLM-based compression consistently achieves remarkable compression rates. *While conventional methods achieve only 3–5× compression, LLMs can reach nearly 23×*. We also conduct an in-depth analysis of the influence of various factors, such as different models, datasets, data sizes, and chunk sizes. Across various model sizes, datasets, data sizes, and chunk sizes, LLM-based compressors maintain significantly higher compression ratios than previous methods.

We hope our findings provide new insights and foundations for the management of text data in large-scale database.

2 Background

In this section, we begin by defining LLM- LLM-generated data and comparing it to conventional machine-generated data, emphasizing their core distinctions. We then examine the key characteristics of LLM-generated text in contrast to other datasets. We summarize the characteristics of different types of data in Table 1. Finally, we delve into the role of randomness in LLM text generation, illustrating how stochastic elements within the model contribute to both diversity and uncertainty in the resulting outputs.

Table 1: Comparison of Machine-Generated Data, LLM-Generated Data, and Human-Generated Data.

Characteristic	Machine-Gen	LLM-Gen	Human-Gen
Automatically Generated	✓	✓	–
Mimics Human Writing	–	✓	✓
High Volume	✓	✓	–
Highly Structured	✓	–	–
Time-Series Nature	✓	–	–
Contains Semantic Context	–	✓	✓
Factual & Deterministic	✓	–	–

2.1 Conventional Machine-Generated Data

Machine-generated data refers to data that is entirely produced by machines or software without human intervention. The generation process is usually automated, occurring at fixed intervals or when triggered by specific events. The core characteristic of machine-generated data is that it originates from computational processes or hardware systems rather than being manually recorded. Examples include system logs, sensor readings, network traffic, and financial records. These data types are automatically generated by computational systems, IoT devices, and network infrastructure, often capturing real-time events and transactions without direct human input. They serve critical roles in monitoring, diagnostics, security analysis, and operational decision-making across industries [53].

2.2 Large Language Model-Generated Data

Synthetic data refers to data that is not derived directly from real-world events but is instead artificially generated by algorithms. LLM-generated synthetic data specifically refers to the text data created by Large Language Models (LLMs). Unlike traditional machine-generated data, which follows predefined rules, LLM-generated data mimics human language patterns, maintains contextual coherence,

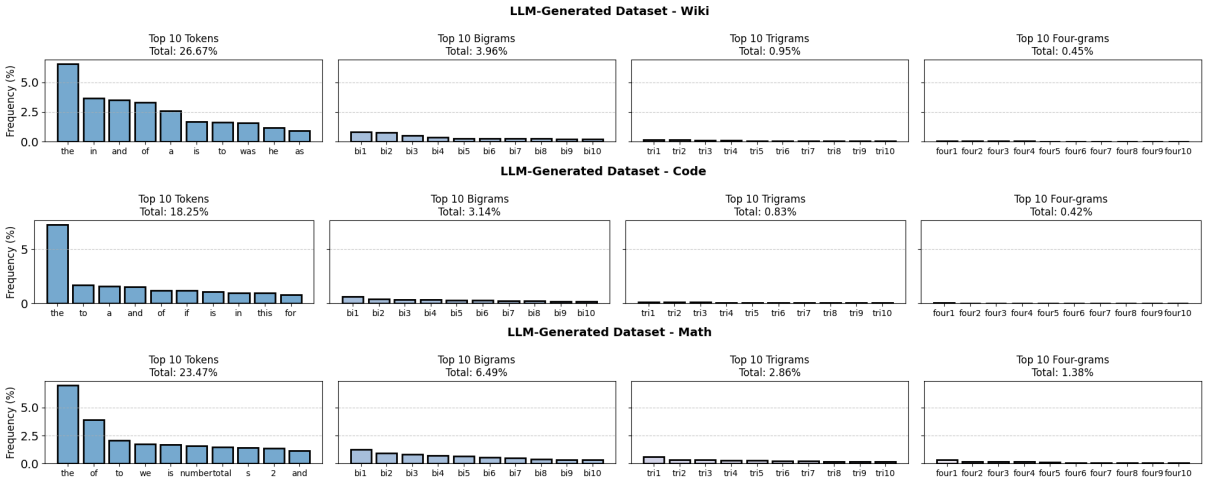

Figure 2: N-gram frequency count for LLM-generated synthetic data.

and exhibits linguistic diversity. Trained on massive corpora, LLMs generate text that closely resembles human writing, making them valuable for applications in content creation, data augmentation, and knowledge synthesis [1, 7, 8, 20, 50].

Code-related datasets dominate with 571 datasets, aligning with the growing importance of code generation and LLMs in software development. Linguistics follows with 471 datasets, reflecting the widespread use of synthetic data in NLP tasks like text generation and translation. Other major categories include art (432 datasets), literature (151 datasets), and music (151 datasets), highlighting AI's role in creative content generation. Science (246 datasets), math (165 datasets), and education (226 datasets) emphasize knowledge-based applications, while medical (131 datasets), business (38 datasets), finance (49 datasets), and law (77 datasets) demonstrate the increasing use of synthetic data in professional and regulatory fields.

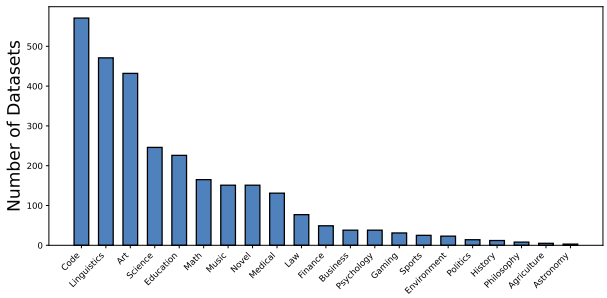

Figure 3: LLM-generated data distribution.

2.3 Characteristics of LLM-Generated Text vs. Other Datasets

We begin by analyzing the characteristics of different types of data, aiming to gain insights that can guide more effective compression strategies. The characteristics of LLM-generated text differ significantly from traditional structured datasets used in database benchmarks. While machine-generated datasets such as TPC-H [11] are designed with well-defined schemas and structured query workloads, LLM-generated text exhibits high variability and redundancy and lacks explicit tabular organization. Similarly, human-generated text data—such as web articles, social media content, and historical corpora—shares some properties with LLM-generated text.

Table 1 compares the key characteristics of machine-generated data, LLM-generated data, and human-generated data. Machine-generated data is highly structured, deterministic, and often time-series-based, whereas LLM-generated data mimics human writing and lacks the strict structure of traditional machine-generated formats. Unlike human-generated data, LLM-generated data is automatically produced and exhibits high volume but may not always be factually accurate.

3 Compressibility Analysis of LLM-Generated Data

In this subsection, we begin by exploring the potential for deduplication in LLM-generated data. We then evaluate the information content of machine-generated, human-generated, and LLM-generated data using entropy on different levels of tokenization and mutual information. Finally, we discuss the limitations of existing compression methods, setting the stage for new strategies that leverage the predictability of LLM-generated data.

3.1 Deduplication Potential of LLM-Generated Data

Deduplication is a common technique for machine-generated data, such as tabular or time-series data [52, 56]. To assess its applicability to LLM-generated data, we performed an n-gram analysis on LLM-generated synthetic datasets to evaluate their redundancy and deduplication potential. By examining the frequency distribution of tokens, bigrams, trigrams, and four-grams across different domains, we aim to obtain a holistic view of the redundancy of these datasets, i.e., how much repeated content exists in these datasets